

Chapter 5.2 - Memory Technologies (SRAM, DRAM, Flash & Disk)

Comprehensive Revision Flashcards • 108 Key Concepts & Practice Questions

#1	Q1: What type of memory is typically used to implement main memory in modern computer hierarchies?
	Answer: DRAM (dynamic random access memory).
#2	Q2: What type of memory is typically used for levels closer to the processor, such as caches?
	Answer: SRAM (static random access memory).
#3	Q3: Which is less costly per bit, DRAM or SRAM?
	Answer: DRAM is less costly per bit than SRAM.
#4	Q4: What is the primary reason for the price difference between DRAM and SRAM?
	Answer: DRAM uses significantly less area per bit, allowing for larger capacity on the same amount of silicon.
#5	Q5: What nonvolatile memory technology is commonly used as secondary memory in Personal Mobile Devices?
	Answer: Flash memory.
#6	Q6: In servers, what technology is used to implement the largest and slowest level in the memory hierarchy?
	Answer: Magnetic disk.
#7	Q7: According to the 2012 data, what is the typical access time for SRAM?
	Answer: 0.5–2.5 nanoseconds.
#8	Q8: According to the 2012 data, what is the typical access time for DRAM?
	Answer: 50–70 nanoseconds.
#9	Q9: According to the 2012 data, what is the typical access time for Flash memory?
	Answer: 5,000–50,000 nanoseconds.

#10	Q10: According to the 2012 data, what is the typical access time for a magnetic disk?
	Answer: 5,000,000–20,000,000 nanoseconds.
#11	Q11: Why is the access time for SRAM very close to its cycle time?
	Answer: Because SRAMs do not need to be refreshed.
#12	Q12: How many transistors per bit do SRAMs typically use?
	Answer: Six to eight transistors per bit.
#13	Q13: What is the primary function of the six to eight transistors in an SRAM cell?
	Answer: To prevent the stored information from being disturbed when read.
#14	Q14: Due to Moore's Law, where are all levels of caches typically integrated today?
	Answer: They are integrated onto the processor chip.
#15	Q15: How is a value stored in a DRAM cell?
	Answer: As a charge in a capacitor.
#16	Q16: How many transistors are used per bit of storage in a DRAM cell?
	Answer: A single transistor.
#17	Q17: Why is DRAM so much denser and cheaper per bit than SRAM?
	Answer: Because DRAMs use only one transistor per bit of storage, compared to SRAM's six to eight.
#18	Q18: Why is DRAM called 'dynamic'?
	Answer: Because the charge stored in its capacitor cells cannot be kept indefinitely and must be periodically refreshed.
#19	Q19: What is the process for refreshing a DRAM cell?
	Answer: Its contents are read and then written back.
#20	Q20: How does the two-level decoding structure in DRAMs facilitate efficient refreshing?
	Answer: It allows an entire row to be refreshed at once with a single read cycle followed by a write cycle.

#21	Q21: In a modern DRAM's internal organization, what command is sent to open or close a bank?
	Answer: A PRE (precharge) command.
#22	Q22: In a modern DRAM, what command causes a row to be transferred to a buffer?
	Answer: An Act (activate) command, which is sent with the row address.
#23	Q23: How does a DRAM improve performance for repeated accesses to the same row?
	Answer: It buffers the entire row, and this buffer acts like an SRAM, allowing for much faster access to bits within that row.
#24	Q24: What is the primary advantage of Synchronous DRAMs (SDRAMs)?
	Answer: The use of a clock eliminates the time needed for the memory and processor to synchronize.
#25	Q25: How do SDRAMs achieve a speed advantage for burst transfers?
	Answer: The clock transfers successive bits in a burst without needing additional address bits to be specified.
#26	Q26: What does DDR stand for in DDR SDRAM?
	Answer: Double Data Rate.
#27	Q27: What is the key mechanism behind Double Data Rate (DDR) technology?
	Answer: Data is transferred on both the rising and falling edges of the clock, effectively doubling the bandwidth.
#28	Q28: A DDR4-3200 DRAM can perform 3200 million transfers per second. What is its clock frequency?
	Answer: 1600 MHz.
#29	Q29: What internal DRAM organization technique allows for simultaneous reads or writes from different sections to increase bandwidth?
	Answer: Organizing the DRAM into multiple banks, each with its own row buffer.
#30	Q30: Term: Address Interleaving
	Answer: A rotating access scheme where an address is sent to several banks, allowing them all to read or write simultaneously to supply higher bandwidth.

#31	Q31: What is a DIMM?
	Answer: A dual inline memory module, which is a small board containing multiple DRAM chips, commonly used for server memory.
#32	Q32: A DIMM using DDR4-3200 SDRAMs, organized to be 8 bytes wide, would be named what, based on its bandwidth?
	Answer: PC25600.
#33	Q33: Term: Memory Rank
	Answer: A subset of chips in a DIMM that share common address lines.
#34	Q34: Flash memory is a type of ____.
	Answer: electrically erasable programmable read-only memory (EEPROM)
#35	Q35: What is a primary limitation of flash memory regarding write operations?
	Answer: Writes can wear out the flash memory bits over time.
#36	Q36: Term: Wear Leveling
	Answer: A technique used by flash memory controllers to spread writes evenly by remapping frequently written blocks to less used blocks.
#37	Q37: What is the main purpose of wear leveling in flash memory products?
	Answer: To prevent premature failure of the memory by distributing the wear from write operations across all blocks.
#38	Q38: What are the rotating components of a magnetic hard disk called?
	Answer: Platters.
#39	Q39: What is the component in a hard disk that contains a small electromagnetic coil to read and write information?
	Answer: A read-write head.
#40	Q40: Term: Track (on a magnetic disk)
	Answer: One of thousands of concentric circles that make up the surface of a magnetic disk.

#41	Q41: Term: Sector (on a magnetic disk)
	Answer: One of the segments that make up a track; it is the smallest amount of information that is read or written on a disk.
#42	Q42: Term: Cylinder (on a magnetic disk)
	Answer: The set of all tracks under the read-write heads at a given point on all surfaces.
#43	Q43: What is the first stage in a disk access process, involving positioning the head over the proper track?
	Answer: A seek.
#44	Q44: Term: Seek Time
	Answer: The time required to move the read-write head to the desired track on a disk.
#45	Q45: Why might the actual average seek time be much lower than the advertised average seek time?
	Answer: Due to the locality of disk references, where successive accesses are often to the same file or nearby tracks.
#46	Q46: Term: Rotational Latency
	Answer: The time required for the desired sector of a disk to rotate under the read-write head.
#47	Q47: What is the average rotational latency generally assumed to be?
	Answer: Half the time of a full rotation.
#48	Q48: What is the third and final component of a disk access time?
	Answer: Transfer time, which is the time to transfer the actual block of bits.
#49	Q49: Transfer time is a function of what three factors?
	Answer: Sector size, rotation speed, and the recording density of a track.
#50	Q50: How do modern high-level disk interfaces organize logical blocks to speed up sequential transfers?
	Answer: They are ordered in a serpentine fashion across a surface, rather than strictly following the old sector-track-cylinder model.

#51	Q51: Compared to DRAM, approximately how much faster is Flash memory?
	Answer: Flash memory is not faster than DRAM; DRAM is approximately 100,000 times faster than a magnetic disk, while Flash is 1000 times faster.
#52	Q52: What is a key advantage of magnetic disks over flash memory regarding data writes?
	Answer: Magnetic disks do not have a write wear-out problem.
#53	Q53: Why is flash memory a better match than magnetic disks for personal mobile devices?
	Answer: Flash memory is much more rugged and can better withstand the jostling inherent in mobile devices.
#54	Q54: In a memory hierarchy, the _____ acts as a safe place to store things that are needed frequently.
	Answer: cache
#55	Q55: Term: Cache
	Answer: Any storage managed to take advantage of locality of access, most commonly the level of memory between the processor and main memory.
#56	Q56: What is a 'cache miss'?
	Answer: An event that occurs when a requested data item is not found in the cache, requiring it to be fetched from a lower level of memory.
#57	Q57: Term: Direct-mapped cache
	Answer: A cache structure in which each memory location is mapped to exactly one location in the cache.
#58	Q58: How is the specific cache location determined for a memory address in a typical direct-mapped cache?
	Answer: By using the formula: (Block address) modulo (Number of blocks in the cache).
#59	Q59: In a direct-mapped cache with a size that is a power of 2, which bits of the address are used as the index?
	Answer: The low-order bits of the address.

#60	Q60: Term: Tag (in a cache)
	Answer: A field in the cache that contains the address information required to identify whether the data in a cache block corresponds to a requested word.
#61	Q61: What part of the memory address is stored in the cache tag?
	Answer: The upper portion of the address, corresponding to the bits that are not used as an index into the cache.
#62	Q62: What is the purpose of a 'valid bit' in a cache?
	Answer: To indicate whether a cache block contains valid information or not (e.g., after a processor starts up).
#63	Q63: What are the four primary technologies used in modern memory hierarchies?
	Answer: DRAM (Dynamic Random Access Memory), SRAM (Static Random Access Memory), Flash memory, and Magnetic Disk.
#64	Q64: In a typical memory hierarchy, what technology is used for main memory?
	Answer: DRAM (Dynamic Random Access Memory).
#65	Q65: In a typical memory hierarchy, what technology is used for cache levels closer to the processor?
	Answer: SRAM (Static Random Access Memory).
#66	Q66: Which is more costly per bit, SRAM or DRAM?
	Answer: SRAM is significantly more costly per bit than DRAM.
#67	Q67: What is the primary reason for the price difference between SRAM and DRAM?
	Answer: DRAM uses significantly less area per bit of memory, allowing for larger capacity on the same amount of silicon.
#68	Q68: What type of memory is typically used as secondary memory in Personal Mobile Devices (PMDs)?
	Answer: Flash memory.
#69	Q69: In servers, what technology is used to implement the largest and slowest level of the memory hierarchy?
	Answer: Magnetic disk.

#70	Q70: What was the typical access time for SRAM in 2012?
	Answer: 0.5 to 2.5 nanoseconds.
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#72	Q72: What was the typical access time for Flash memory in 2012?
	Answer: 5,000 to 50,000 nanoseconds.
#73	Q73: What was the typical access time for a magnetic disk in 2012?
	Answer: 5,000,000 to 20,000,000 nanoseconds.
#74	Q74: _____s have a fixed access time to any datum, though the read and write access times may differ.
	Answer: SRAM
#75	Q75: How many transistors do SRAMs typically use per bit of storage?
	Answer: Six to eight transistors per bit.
#76	Q76: Why do SRAMs not need to be refreshed?
	Answer: As long as power is applied, the value can be kept indefinitely without the charge leaking away.
#77	Q77: What is the primary modern application for SRAM technology?
	Answer: All levels of caches integrated directly onto the processor chip.
#78	Q78: How is a value stored in a Dynamic RAM (DRAM) cell?
	Answer: As a charge in a capacitor.
#79	Q79: The need for a DRAM cell to be periodically _____ is why this memory structure is called dynamic.
	Answer: refreshed
#80	Q80: How does DRAM's two-level decoding structure make refreshing efficient?
	Answer: It allows an entire row to be refreshed at once with a single read and write cycle.

#81	Q81: In a modern DRAM, what does sending an Act (activate) command with a row address accomplish?
	Answer: It causes the entire specified row to be transferred to a buffer.
#82	Q82: How do DRAMs improve performance for repeated accesses to the same row?
	Answer: They buffer the entire row, and this buffer acts like a faster SRAM for subsequent accesses within that row.
#83	Q83: What is the main advantage of Synchronous DRAMs (SDRAMs)?
	Answer: The use of a clock eliminates the time needed for the memory and processor to synchronize.
#84	Q84: What does the name 'Double Data Rate (DDR) SDRAM' signify?
	Answer: Data is transferred on both the rising and falling edges of the clock, doubling the bandwidth for a given clock rate.
#85	Q85: A DDR4-3200 DRAM can perform 3200 million transfers per second. What is its clock rate?
	Answer: 1600 MHz.
#86	Q86: What is the name of the technique that uses multiple internal DRAM banks to increase bandwidth by rotating accesses between them?
	Answer: Address interleaving.
#87	Q87: Term: DIMM
	Answer: Definition: Dual Inline Memory Module, a small board containing multiple DRAM chips, commonly used for server memory.
#88	Q88: How is the name of a DIMM, such as PC25600, derived?
	Answer: It is named after its bandwidth in megabytes per second (e.g., a DDR4-3200 DIMM that is 8 bytes wide has a bandwidth of 25,600 MB/s).
#89	Q89: What term is used for the subset of chips on a DIMM that share common address lines and are used for a particular transfer?
	Answer: Memory rank.

#90	Q90: Flash memory is a type of _____ memory.
	Answer: electrically erasable programmable read-only memory (EEPROM)
#91	Q91: What is the primary durability issue with flash memory?
	Answer: Writes can wear out the memory bits over time.
#92	Q92: What is the name of the technique used in flash memory to spread writes across different blocks to prevent premature wear?
	Answer: Wear leveling.
#93	Q93: How does wear leveling work in flash memory controllers?
	Answer: It remaps blocks that have been written many times to less frequently used blocks.
#94	Q94: A magnetic hard disk consists of a collection of _____, which rotate on a spindle.
	Answer: platters
#95	Q95: What is the name of the small electromagnetic coil on a movable arm used to read and write information on a hard disk?
	Answer: A read-write head.
#96	Q96: What is a 'cylinder' in the context of a magnetic disk?
	Answer: The set of all tracks that are under the read-write heads at a given point on all surfaces.
#97	Q97: What are the three main stages of a disk access process?
	Answer: Seek time, rotational latency, and transfer time.
#98	Q98: Term: Seek (in disk access)
	Answer: Definition: The process of positioning a read/write head over the proper track on a disk.
#99	Q99: The time required to move the disk head to the desired track is called the _____ time.
	Answer: seek

#10 0	Q100: On average, how much of a full circle must a disk platter spin to reach the desired sector?
	Answer: Half a rotation.
#10 1	Q101: What is the final component of a disk access, which is a function of sector size, rotation speed, and recording density?
	Answer: Transfer time.
#10 2	Q102: What is a primary difference between magnetic disks and semiconductor memory like DRAM in terms of access time?
	Answer: Disks have a much slower access time because they are mechanical devices (DRAM is ~100,000 times faster).
#10 3	Q103: What is a primary difference between magnetic disks and semiconductor memory like DRAM in terms of cost?
	Answer: Disks are significantly cheaper per bit (10 to 100 times cheaper).
#10 4	Q104: Unlike flash memory, magnetic disks do not have a _____ problem.
	Answer: write wear-out
#10 5	Q105: Why is flash memory a better match for personal mobile devices than magnetic disks?
	Answer: Flash is much more rugged and can better withstand the jostling inherent in mobile devices.
#10 6	Q106: What is a 'direct-mapped cache'?
	Answer: A cache structure in which each memory location is mapped to exactly one location in the cache.
#10 7	Q107: How is the cache location for a memory address typically determined in a direct-mapped cache?
	Answer: By using the formula: (Block address) modulo (Number of blocks in the cache).
#10 8	Q108: In a cache, what is the purpose of a 'tag'?
	Answer: It contains the address information required to identify whether the data in a cache block corresponds to the requested word.